

REVIEW

2025 Donald Seldin Lecture: Leveraging Diverse Population Genomics and Multiomics Integration for Gene Discovery of Cardiovascular and Kidney Diseases

Andrea R.V.R. Horimoto¹ , Nora Franceschini² 

ABSTRACT: This review discusses the implications of frameworks leveraging genetic admixture and multiomics data for advancing gene discovery in cardiovascular and kidney disease research. By broadening gene discovery efforts to additional populations that have a disproportionately high risk of disease and leveraging genetic diversity in admixed populations, studies can identify population-enriched risk variants that traditionally have been missed in genome-wide association studies. The use of multiomics approaches, including the transcriptome, proteome, and metabolome, advances a mechanistic understanding of disease beyond associations. As single-cell omics technologies continue to improve, their integration into gene discovery may help uncover cell-type-specific regulatory pathways and more precise biological contexts. The full potential of these approaches depends on sustained investment in diverse, well-characterized omics data sets, methodological innovation in multiancestry statistical approaches, and interdisciplinary collaboration bridging genomics, epidemiology, and clinical medicine. These efforts will need to be translated into clinically actionable insights, including ancestry-informed risk stratification and targeted therapeutics, to improve outcomes for cardiovascular and kidney diseases.

Key Words: clinical medicine ■ genetic variation ■ hypertension ■ kidney ■ multiomics ■ proteome ■ risk assessment

Several chronic conditions commonly seen in clinical care have shown significant disparities in disease burden across populations, including United States (US) racial and ethnic groups.^{1,2} Hypertension is more prevalent and often more severe in US Black individuals, who also face a high burden of chronic kidney disease (CKD) and cardiovascular disease.³ Hispanic and Latino individuals have high rates of metabolic diseases, such as type 2 diabetes, which are known causes of cardiovascular and kidney diseases.^{4,5} These differences in disease burden are likely due to the interplay of genetic susceptibility, socioeconomic factors (including access to care), and behavior and lifestyle factors, which vary across racial and ethnic groups. Studies leveraging genetic diversity can provide further understanding of differences in disease occurrence and risk, and insights into disease pathogenesis and therapy.⁶

Leveraging genetic diversity is also important for gene mapping to understand disease causes and risk contexts,

and to support clinical applications of genetic data in clinical care. Genomic data have potential applications for screening individuals at risk, the diagnosis of disease subtypes, and the prediction of outcomes (eg, drug toxicity or efficacy, or disease complications). The limited genetic research in high-risk groups remains a critical barrier to the implementation of genetic data in disease prevention, diagnosis, and prognosis across populations. Because populations differ in their genetic architecture, including the presence of ancestry-enriched disease-risk variants and the distinct linkage disequilibrium structures, genomic findings cannot be directly or reliably transferred across populations.⁷ This research gap is relevant to the US population, given its large genetic diversity and admixture.⁸

Recent research has combined multiomics data to improve gene discovery. New approaches that query gene transcripts, such as transcriptome-wide association studies (TWASs), use predicted gene expression

Correspondence to: Nora Franceschini, MD, MPH, University of North Carolina, 123 W Franklin St, Ste 410, Chapel Hill, NC 27516. Email noraf@unc.edu
© 2026 American Heart Association, Inc.

Hypertension is available at www.ahajournals.org/journal/hyp

Nonstandard Abbreviations and Acronyms

CKD	chronic kidney disease
eQTL	expression quantitative trait loci
GWAS	genome-wide association study
HCHS/SOL	Hispanic Community Health Study/ Study of Latinos
RNA-seq	RNA sequencing
SNP	single nucleotide polymorphism
TWAS	transcriptome-wide association study

as a molecular phenotype for gene discovery. These approaches have shown gains in gene discovery and can be extended to other molecular data, including tissue- and cell-specific omics, for further biological insights. Therefore, multiomics approaches can advance gene discovery in populations.

This review focuses on gene discovery approaches beyond genome-wide association studies (GWASs), including approaches that leverage population admixture, such as admixture mapping. We additionally discuss emerging approaches that leverage tissue- and single-cell omics to identify disease-associated genes, proteins, and metabolites in target tissues and cells. We emphasize gaps in research, including the need to extend these omics studies to high risk, diverse populations. Key genetic definitions are included in Table 1. We provide examples of these approaches using cardiovascular and CKD traits and include a section on hypertension to describe available studies.

GENETIC RESEARCH LEVERAGING ANCESTRY FOR GENE DISCOVERY

Studies of genetic susceptibility have traditionally been conducted using GWAS approaches for clinical diseases and traits.⁹ GWAS discovery relies on common variants with small effects, and large samples are needed to achieve sufficient power to detect these associations.

Gene discovery is often biased toward populations that are overrepresented in GWAS. Based on the GWAS Diversity Monitor, the majority of studies (87.8%) have been conducted in individuals of European ancestry.⁸ Admixed populations, such as Black (3.7%) and Hispanic or Latino (1.7%) populations, and other worldwide populations remain markedly underrepresented in genetic studies.⁸

Genome-wide gene mapping approaches that leverage genetic ancestry can yield additional gains in gene discovery beyond those from GWAS in admixed populations. Genetic ancestry is the population genetic background shaped by human history and migration and is estimated by comparing an individual's genome-wide genetic variation with reference genetic maps of populations in distinct worldwide geographic regions. Admixture mapping is an especially useful approach for diseases that have shown significant differences in prevalence across populations, such as hypertension and CKD. However, admixture mapping approaches have been underused in gene discovery.

GENETIC DIVERSITY, GLOBAL AND LOCAL ANCESTRY ADMIXTURE

Human genetic diversity results from demographic events and natural selection in response to specific environmental contexts.¹⁰ Populations differ in allele frequencies of variants and in the correlations among variants (linkage disequilibrium). Some variants that confer disease risk may be present in some populations and not others. Examples include a risk haplotype in the *SLC16A11* gene, which was first identified for association with type 2 diabetes risk in Mexican individuals and some Latin American groups. The haplotype has a low frequency in East Asians (10%) and is rare in individuals of European and African ancestry.¹¹ Some population-enriched variants may result from adaptations to environmental exposures, such as diet and infections, as well as to the physical environment.^{10,12,13} For example, variants in the *FADS* (fatty acid desaturase) gene cluster identified in

Table 1. Key Definitions

Ancestry	The population genetic background designed by human history and migration. It is estimated by comparing an individual's genome-wide genetic variations to reference genetic maps of ancestral populations across different geographic regions worldwide. This refers exclusively to the individual's genetic background, not race or ethnicity, which are social constructs as previously discussed.
Ancestral populations	Populations that contributed genetically to the genomes of present-day individuals. Typically characterized by distinct allele frequency patterns shaped by their evolutionary and geographic histories.
Cis-eQTL	The expression quantitative trait locus (eQTL) that regulates nearby genes; it is usually located within 1 megabase from the gene transcription site.
Global ancestry	Proportions of each contributing ancestral population, averaged across the entire genome.
Local ancestry	Ancestral origin of chromosomal segments (haplotypes) of an individual's genome.
Locus	A specific chromosomal region, identified by its physical position within the chromosome.
Trans-eQTL	The expression quantitative trait locus (eQTL) that regulates the expression of genes far away (over 1 megabase from the gene transcription site) or on different chromosomes.

Greenland Inuit are related to physiological adaptations to a diet rich in polyunsaturated fatty acids,¹³ and variants in several genes have been related to physiological adaptations to high-altitude hypoxia in Andean populations.¹² Studies have identified African ancestry variants resulting from adaptation to endemic pathogens in Africa.^{6,10,14–17} This includes the *APOL1* (Apolipoprotein L1) risk variants (G1, rs73885319 [p.S342G] and rs60910145 [p.I384M] missense mutations in complete linkage disequilibrium; G2, rs71785313, a 6-bp deletion [p.delN388/Y389]) that were selected for protection against sleeping sickness infection, and *HBB* (hemoglobin subunit β ; related to sickle cell disease, rs334, missense mutation [p.Glu6Val]), *G6PD* (glucose-6-phosphate dehydrogenase), and *DARC* (Duffy antigen/chemokine receptor) variants selected for protection against malaria.^{6,10} Both *APOL1* G1/G2 variants and 1 copy of the *HBB* rs334 variant related to sickle cell trait are associated with CKD in Black and Hispanic and Latino individuals.^{6,14,15} *G6PD* deficiency variants are associated with lower hemoglobin levels and hemoglobin A1c, contributing to disparities in diabetes outcomes in US Black and African individuals, and the null *DARC* variant is associated with lower white blood cell counts in these populations.^{16,17} These variants are selectively present in some populations, and individuals with admixed ancestry can carry these risk variants. Populations with less-characterized genomes may harbor additional population-enriched disease-risk variants. Importantly, these risk variants can only be detected when these populations are included in gene discovery.

Individuals with ancestry admixture have chromosomes that are mosaics of chromosome segments of different population ancestral origins that can be mapped to different populations (local ancestry). Averaging the local ancestry proportions provides a measure of the global ancestry for each contributing ancestral population. In general, US Black individuals have ancestry admixture of African and European ancestries, while US Hispanic and Latino individuals have ancestry admixture from African, European, and Native American ancestries (Figure 1A). Our studies in the Hispanic Community Health Study/Study of Latinos (HCHS/SOL) have identified varying African, European, and Native American global ancestry proportions among individuals reporting their country of origin or US region as Mainland (Mexico, Central and South America) or the Caribbean (Dominican Republic, Cuba, Puerto Rico; Figure 1B). Caribbean participants had, on average, 24% global African ancestry, while Mainland participants had 5% to 15% global African ancestry.¹⁴ In comparison, US Black individuals have $\approx$ 75% to 80% global African ancestry.

Global ancestry may not capture disease risk, as shown in a secondary analysis of the Systolic Blood Pressure Reduction Intervention Trial, which did not find associations between West African global ancestry and

trajectories of blood pressure and kidney function.¹⁸ Local ancestry can inform ancestry risk alleles and gene discovery. In individuals with ancestry admixture, causal variants are expected to occur more frequently on chromosomal segments inherited from ancestral populations with a higher disease burden. Therefore, genetic risk variants enriched in a population of origin may segregate in individuals with admixture and confer unrecognized disease risk. For example, in the HCHS/SOL Caribbean participants, we identified African ancestry risk variants associated with CKD. The *APOL1* high risk (G1/G2 2 copies) and the *HBB* variant related to sickle cell trait (1 copy) were present in 1% and 2% of participants, respectively.¹⁴ This is in comparison with US Black individuals, for whom the prevalences of these variants are 10% to 15% and 8% to 10%, respectively. We also showed that *APOL1* G1 and G2 risk alleles (2 versus 0/1 copies) and the *HBB* rs334 variant (1 versus 0 copies) were significantly associated with 2.9-fold (95% CI, 1.5–5.4) and 2.4-fold (95% CI, 1.7–3.5) higher odds of CKD, respectively, among HCHS/SOL Caribbean participants.¹⁴ These findings support the increased risk of ancestry-specific variants in admixed populations, and they have clinical implications for CKD diagnosis and potential screening of kidney donors in groups not traditionally considered at risk, especially for *APOL1* G1 and G2 variants.¹⁹

ADMIXTURE MAPPING VERSUS GWAS

GWAS tests the association of single-nucleotide polymorphisms (SNPs) across the genome with traits. Multiple GWAS data sets are combined through meta-analysis, increasing sample size and statistical power to detect associations. Recent studies have included multiple populations to represent genetic diversity, improving precision in identifying loci and localizing disease-causing variants. Large, combined samples have also allowed for querying low-frequency variants (minor allele frequency <1%). However, GWAS does not specifically address admixture or ancestry-enriched loci or variants. Instead, estimated global ancestry is used only to account for differences in population structure across populations (eg, differences in allele frequencies).

Admixture mapping approaches leverage ancestry-specific genomic regions (or haplotypes) to identify causal variants enriched in their population of origin for both rare and common variants.²⁰ Admixture mapping tests the counts of ancestry-specific haplotypes (local ancestry) across the genome for association with an outcome and has higher statistical power to detect associations given a lower number of tests performed. Consequently, the sample sizes needed for gene discovery in admixture mapping are smaller than those of GWAS. A comparison between admixture mapping and GWAS approaches is shown in Figure 2A. Like GWAS, admixture mapping

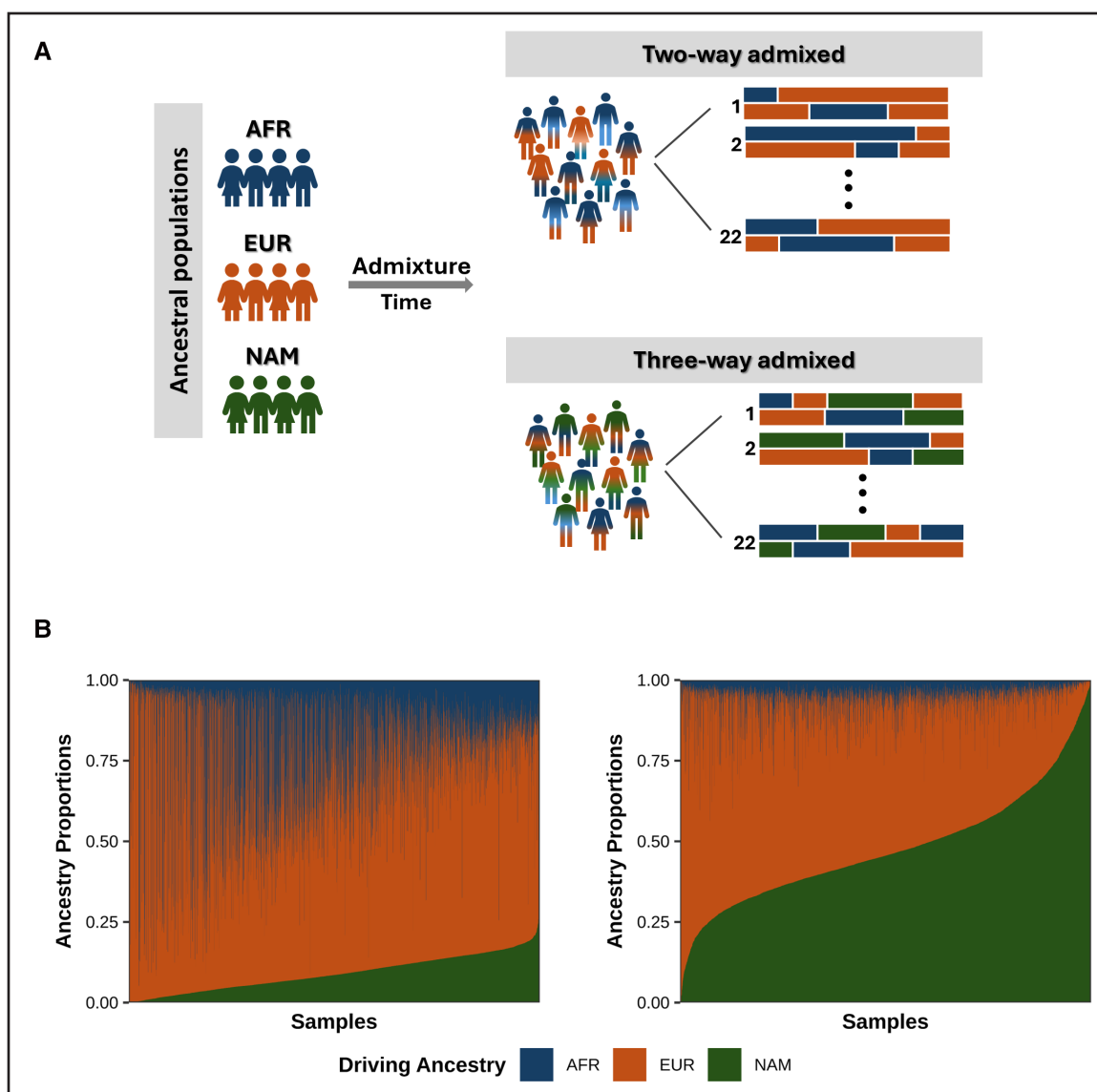


Figure 1. Admixture and diversity of multiancestry populations.

A, Representation of populations with 2-way (eg, US Black) and 3-way ancestry admixture (eg, US Hispanic and Latino), showing African (AFR), European (EUR), and Native American (NAM) ancestral populations. Recent admixture generates extended haplotypes from multiple ancestral sources, resulting in genomes that are mosaics of chromosomal segments with distinct ancestral origins. **B**, Global ancestry proportions of Hispanic Community Health Study/Study of Latinos (HCHS/SOL) Caribbean (at left) and Mainland (at right) participants. Vertical bars show the ancestry proportions for each individual, arranged in ascending order of estimated NAM ancestry.

requires further analysis to fine-map genomic regions to identify the genetic variants that drive the associations.²² While GWASs usually query common variants, admixture mapping findings are expected to have the contribution of one or more variants across the full spectrum of allele frequencies.

Admixture mapping effectiveness relies on substantial allele frequency differences between ancestral populations, reducing sensitivity when these differences are small. Accurate inference of local ancestry and access to well-characterized reference panels for ancestral populations are crucial, as errors can reduce power or introduce bias. This is a challenge for the analysis of admixed genomes, which include understudied ancestral

populations, such as Native American ancestry. Comprehensive public data sets reflecting the diversity of populations remain limited by small samples, which support the need for increased representation in genetic studies.

GENETIC INSIGHTS FROM ADMIXTURE MAPPING STUDIES

African ancestry *APOL1* locus and CKD

The *APOL1* risk variants G1 and G2 on chromosome 22 were initially identified through admixture mapping of US Black individuals in 2 independent studies that used 190 individuals with CKD due to focal

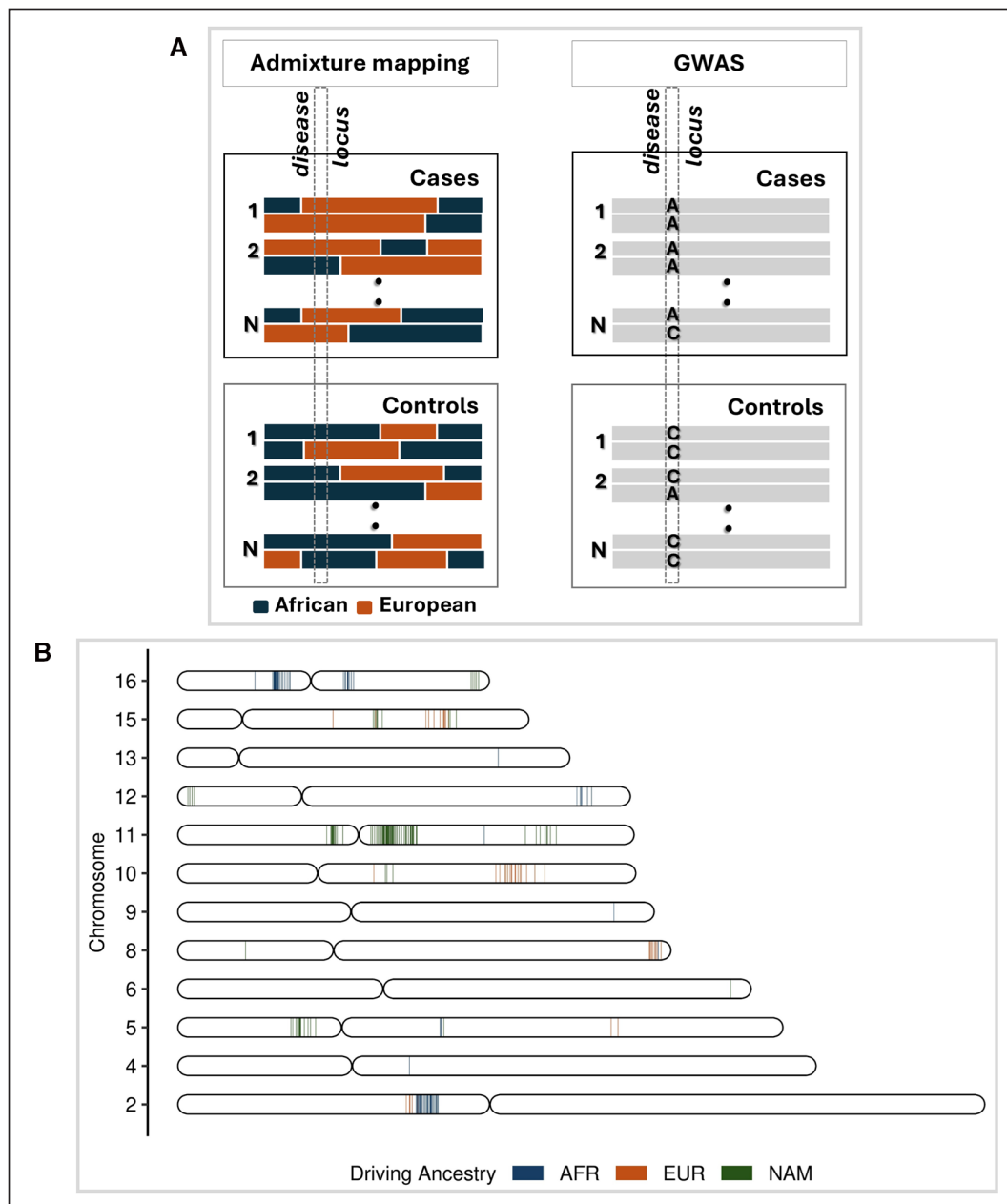


Figure 2. Admixture mapping and genome-wide association study (GWAS) for gene discovery.

A, Comparison of admixture mapping and GWAS approaches. Admixture mapping tests associations between local ancestry (the number of ancestry-specific haplotypes at a given locus) and traits, whereas GWAS tests the association between the reference SNP allele dosage and traits. **B**, Ancestry-driven associations for metabolites from admixture mapping conducted in Hispanic Community Health Study/Study of Latinos (HCHS/SOL) participants. Figure 2B was adapted from Reynolds et al.²¹

segmental glomerulosclerosis and 222 controls,²³ and 1372 hypertension-related end-stage kidney disease cases and 806 controls.²⁴ The locus was subsequently fine-mapped to the *APOL1* gene for gain-of-function alleles that follow a recessive pattern of inheritance.⁶ *APOL1* G1 and G2 risk variants are associated with infection- and immune-related CKD, such as HIV-associated nephropathy and COVID-associated nephropathy.^{19,25} The locus has additional genetic variants that modulate CKD risk, and new therapies are currently under development to target *APOL1* channel

activity or gene function using small molecule inhibitors, antisense oligonucleotides, or by blocking proinflammatory JAK/STAT (Janus kinase/signal transducer and activator of transcription) pathways.^{26,27} *APOL1* risk-variant discovery exemplifies how these approaches can help to uncover novel causal mechanisms of CKD and drive new strategies for disease treatment and prevention. The *APOL1* risk variants remain one of the strongest genetic risk factors for CKD in individuals with African ancestry admixture. Importantly, admixture mapping identified the *APOL1* locus in small case-control studies that included

individuals with African ancestry, and large GWASs of CKD that do not include individuals with African ancestry are not able to detect this locus, given that the variants are monomorphic in other populations.

Native American ancestry loci and long-chain polyunsaturated fatty acid metabolites

Metabolites are the end products of metabolic cellular pathways that are influenced by genetic and environmental factors, such as diet. Prior research has identified Native American ancestry haplotypes contributing to the fatty acid desaturase *n*-3 biosynthetic pathway associated with long-chain-polyunsaturated fatty acid metabolism that are under selection for high dietary fatty acid intake.²⁸ Our admixture mapping studies of metabolites in 3887 HCHS/SOL Hispanic and Latino participants identified significant ancestry-driven associations for 78 circulating metabolites at 484 independent regions (232 novel), including 2 Native American-enriched genomic regions on chromosomes 11 and 15 associated with lipid metabolites, mapping to *FADS1/FADS2* and *LIPC* (hepatic lipase), respectively (Figure 2B).²¹ The chromosome 11 region was associated with a wide range of lipid-based metabolites related to long-chain polyunsaturated fatty acid, with reduced metabolites in both the *n*-3 (eicosapentaenoic acid and docosahexaenoic acid) and *n*-6 (arachidonic acid) biosynthetic pathways, which are precursors to metabolites that impact biological processes related to inflammation or its resolution, and physiological functions (eg, prostaglandins, thromboxanes, leukotrienes, resolvins). These findings support ancestry-driven genetic variation influencing metabolic pathways, which may have implications for cardiovascular, kidney, and immune diseases, as well as for dietary recommendations.²⁹ *FADS1/FADS2* genetic variation is associated with cardiometabolic traits,²⁸ but its impact on cardiovascular and kidney outcomes will need to be assessed in the context of dietary intake, ancestry variation, and other nongenetic risk factors.^{30,31}

OMICS-BASED DISCOVERY

TWAS and Multiomics Approaches

Genome-wide association approaches have been successful in identifying genetic variants linked to complex traits and diseases, with 625,113 SNPs reported in the GWAS Catalog for association with over 15,500 traits by 2024.⁹ However, because associated variants may not directly implicate causal genes, the biological interpretation of GWAS signals remains challenging. Recent advances in multiomics technologies have enabled integrative approaches that directly connect genetic variation to biological function by linking GWAS loci to gene expression, protein abundance, and downstream

molecular pathways, thereby enhancing our understanding of the causal mechanisms of complex diseases (Figure 3A). For example, multiomics integration of GWAS findings with transcriptomic and proteomic data has linked coronary artery disease-associated variants at the *BCAR1* locus to altered vascular gene expression,³² and similar approaches applied to heart failure have identified genetically regulated circulating proteins.³³ Together, these studies suggest that disease-associated variants act through coordinated effects on gene expression and protein abundance to influence cardiovascular pathophysiology.

Many GWAS-identified associations are from variants in noncoding regions, which can influence traits by regulating gene expression (or the transcriptome).⁹ Gene expression data can be leveraged through 2 complementary approaches: expression quantitative trait locus (eQTL) studies, which map genetic variants that influence gene expression, and TWAS, which incorporates eQTL-derived weights to identify gene-trait associations across the genome. Genome-wide eQTL studies complement GWAS by identifying the functional targets of noncoding variants in an unbiased, genome-wide manner, without requiring prior knowledge of specific regulatory regions. To date, these studies have been predominantly conducted using gene expression derived from bulk tissues.³⁴ For example, an eQTL study of participants from the Framingham Heart Study identified over 6 million significant *cis*-eQTL-gene transcript pairs and ≈ 1.5 million *trans*-eQTL-gene transcript pairs, with enrichment for SNPs associated with cardiovascular risk factors.³⁵ However, eQTL studies require access to individual-level genetic and gene expression data, which limits their broad application due to data access restrictions and privacy concerns. TWAS was developed to extend this framework by incorporating genetically predicted gene expression data, thereby enabling analysis in large cohorts where expression data are unavailable.^{36,37}

TWAS approaches have facilitated the identification of novel candidate genes associated with disease susceptibility by integrating genetic variation and gene expression data into a unified framework that tests gene-level associations with a trait.³⁸ Specifically, TWAS connects the functional role of noncoding variants identified in GWAS to genes. It directly integrates genetic data and the gene expression predicted from reference panels to identify candidate genes whose expression is genetically regulated and associated with the trait. Several TWAS approaches have been developed since 2015, given the availability of blood and tissue RNA-sequencing (RNA-seq) data, allowing for the use of gene expression derived from a single or multiple tissues. Additional developments rely on GWAS summary statistics results,³⁹ addressing the limitations of individual-level data.

More recent approaches have further extended these frameworks by enabling the integration of eQTLs that

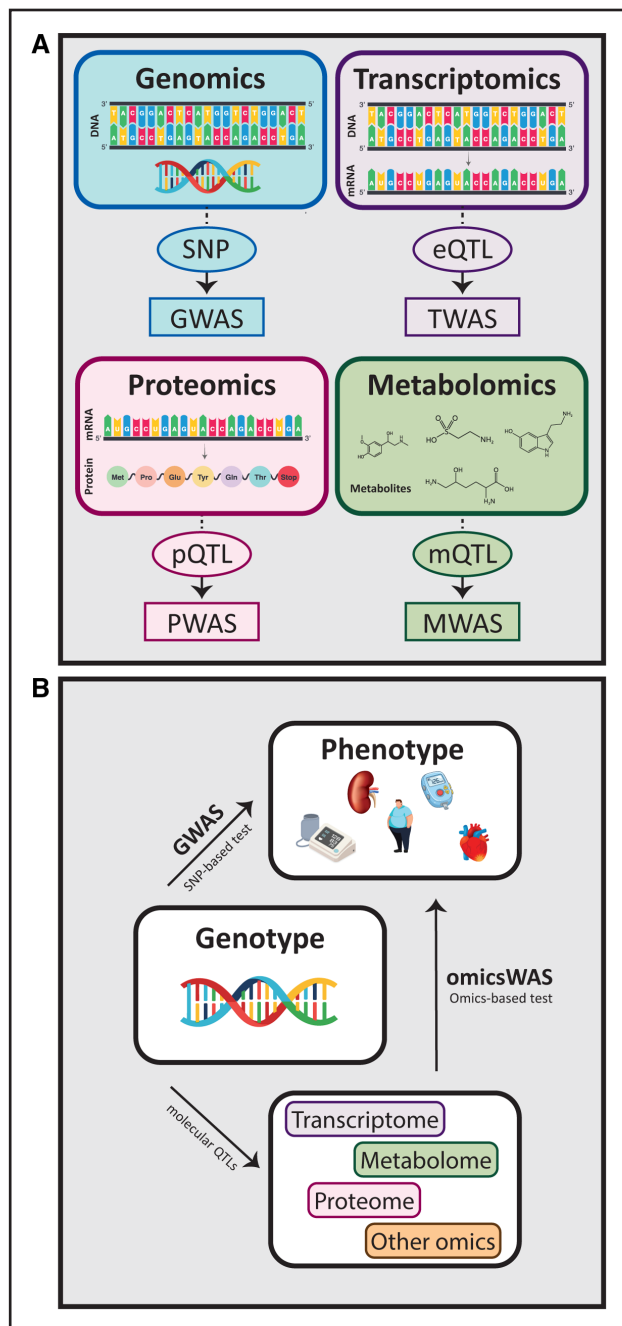


Figure 3. Integrative analytical frameworks combining genetic data with molecular- quantitative trait loci (QTLs).

A, The multiomics cascade and the associated molecular QTLs used to conduct genome-wide association study (GWAS), transcriptome-wide (TWAS), proteome-wide (PWAS), and metabolome-wide (MWAS) association studies. **B**, Analytical frameworks to perform omics-wide association studies of traits.

regulate nearby genes (*cis*-eQTLs, usually within one megabase of the transcription site) and those that regulate the expression of genes far away or on different chromosomes (*trans*-eQTLs).^{36,37} For example, Omnibus Transcriptome Test Using Expression Reference Summary Data³⁷ is a 2-stage approach that offers high flexibility, as the eQTL panels derived in the first stage can

be readily applied across multiple independent association studies. Almost all protein-coding genes have at least 1 *cis* variant associated with expression levels (*cis*-eQTL) or a common variant affecting alternative splicing (*cis*-sQTL). *Cis*-sQTLs are available in blood-based data in the Trans-Omics for Precision Medicine⁴⁰ and multi-tissue data from the Genotype-Tissue Expression database.⁴¹

Most TWAS studies to date, however, rely on bulk-tissue RNA-seq data, which lack the resolution to capture the contributions of specific cell types to disease pathogenesis. Single-cell RNA-seq and single-nucleus RNA-seq are high-resolution methods that enable cell-type-specific analyses, thereby enhancing precision in TWAS gene discovery.⁴² While single-cell RNA-seq offers a more comprehensive view of the transcriptome by sequencing the entire cell, including mitochondrial DNA, rather than only nuclei as in single-nucleus RNA-seq, it has some methodological constraints, such as cell dissociation issues and a restriction to fresh tissues, that are overcome by single-nucleus RNA-seq technologies.

Leveraging complementary omics layers, including the genome, transcriptome, proteome, and metabolome (Figure 3B), can enhance gene discovery by capturing biological variation across multiple levels of regulation. While TWAS specifically leverages the transcriptome, which encompasses RNA molecules produced during the initial stage of gene expression (transcription), gene expression is a multistage process in which post-transcriptional events play a key role in shaping tissue-specific proteomes in humans.⁴³ This has motivated the development of frameworks that operate on other omics layers, each capturing a distinct stage of the biological pathways from genotype to phenotype.

Similar to eQTL mapping underlying TWAS, genome-wide analyses can associate genetic variants with protein levels (pQTLs), metabolite levels (mQTLs), and DNA methylation at specific CpG sites (meQTLs), among other molecular phenotypes. Proteome-wide association studies extend the TWAS paradigm to the proteome by connecting genes to diseases through the direct effects of genetic variants on protein expression (pQTLs),⁴⁴ capturing posttranscriptional regulation that TWAS alone cannot resolve. Similarly, metabolome-wide association studies identify genes whose metabolite levels are genetically regulated (mQTLs) and are associated with diseases and traits. The metabolome, comprising the set of endogenous and exogenous low-molecular-weight compounds (metabolites) involved in cellular metabolism, represents the downstream product of complex interactions among genes and proteins, offering a deeper layer of insight into the molecular pathways implicated in disease. Importantly, analytical pipelines originally developed for TWAS, such as the Omnibus Transcriptome Test Using Expression Reference Summary Data framework, can be readily extended to incorporate other molecular

QTLs, such as pQTLs and mQTLs, enabling truly omics-wide association studies.³⁷ These frameworks can be further adapted to single-cell omics data to yield cell-type-specific insights into disease mechanisms.

Emerging technologies, such as spatial transcriptomics, add another dimension by enabling the in situ quantification of transcripts and proteins, preserving spatial context within tissues and offering new opportunities to link genetic variation to spatially resolved molecular phenotypes. Computational methods and limitations of spatial multiomics applied to cardiovascular diseases have been discussed elsewhere.⁴⁵ Spatial transcriptomics has already been applied to generate integrative maps of human cardiac remodeling after myocardial infarction,⁴⁶ to dissect proinflammatory microvascular changes in ischemic heart disease,⁴⁷ and to characterize spatial glomerular transcriptomic signatures in CKD,⁴⁸ collectively underscoring its potential to reveal cell-type-specific and spatially contextualized mechanisms that remain invisible to bulk-based approaches.

Multiomics Data Are Limited in Diverse Populations

Although advances in high-throughput technologies have enabled multiomics studies, the availability of molecular QTLs in diverse populations and across tissues and cell types remains limited. To the best of our knowledge, no existing tool systematically monitors the representation of diverse populations across association studies using some omics data. Studies have shown the importance of using eQTL panels that more closely match the genetic ancestry of the GWAS data set for better performance of TWAS.^{49,50} It is therefore reasonable to expect that the underrepresentation of omics data in some populations will influence the performance of these studies, as observed in GWAS. Approaches using local ancestry for TWAS have been developed and could be extended to other pathway-level inferences.⁵¹

While multiomics data have been extensively collected from individuals of European ancestry, various initiatives have also worked to diversify population representation. For example, the Genotype-Tissue Expression Consortium has made large-scale bulk-tissue *cis*-eQTL data publicly available, including donors of African ancestry,⁴¹ and the International Human Epigenome Consortium has some diverse-population epigenomics samples.⁵² Other studies focusing on Black, Hispanic and Latino individuals have also reported eQTLs derived from whole-blood bulk tissue.⁵³ More recently, additional consortia have generated and released single-cell eQTL resources across various tissues and cell types by aggregating data from multiple studies.^{54–56} However, they have mostly centered on European ancestry data.

Similarly, a few studies focused on admixed populations have made pQTLs and mQTLs publicly available.^{57,58}

The National Heart, Lung and Blood Institute Trans-Omics for Precision Medicine program is generating whole-genome sequencing, RNA-seq, proteomics, metabolomics, and DNA methylation in a large number of individuals from diverse populations.⁵⁹ Resources from Trans-Omics for Precision Medicine are available in BioData Catalyst and the Database of Genotypes and Phenotypes, and are invaluable for studies focused on admixed populations. Collectively, these efforts represent important progress. Table 2 lists some selected examples of multiomics resources relevant to cardiovascular and kidney research and their ancestry representation.

There is still underrepresentation of admixed Black, Hispanic and Latino, and other groups across multiomics layers and in single-cell atlases underscoring a critical need for large-scale, ancestry-diverse data generation to ensure biological discovery and translational impact.^{50,68,69} This limits the discovery of ancestry-specific regulatory mechanisms⁷⁰ and may bias the interpretation of cell-type-resolved disease mechanisms in cardiovascular and kidney diseases. Addressing this gap through the inclusion of more populations in omics studies and expanding representative reference data sets is essential for biologically robust gene discovery.

Insights From Multiomics Studies for Hypertension in US Admixed Individuals

Large initiatives towards Black, Hispanic and Latino individuals, including the Continental Origins and Genetic Epidemiology Network consortium,⁷¹ the HCHS/SOL,⁷² the Trans-Omics for Precision Medicine Program,⁵⁹ the All of Us Program,⁷³ and several biobanks,⁷⁴ have generated genomic (array or whole-genome sequencing) data in admixed ancestry groups,⁷³ facilitating the investigation of genetic variants associated with hypertension through GWAS,^{75,76} admixture mapping,^{77–80} and polygenic risk score studies.^{81,82} Additional available results from large multi-population gene-by-environmental interaction GWASs of blood pressure traits identified genetic loci that interact with lifestyle (alcohol intake and smoking),^{83,84} psychosocial factors⁸⁵ and sleep patterns.⁸⁶

Individual-level genotypes or GWAS data for hypertension can be leveraged to conduct TWAS, proteome-wide association studies, metabolome-wide association studies, or association studies involving other molecular QTLs. Several TWASs for hypertension have been published using tissues relevant to hypertension, such as the left ventricle, aorta, and whole blood, and across tissues, but all were conducted in European ancestry. For example, TWAS findings highlighted 15 novel genes associated with hypertension in the left ventricle, aorta, whole blood, and peripheral blood, including *RBM6*, *HLA-DRB5*, *UHRF1BP1*, *LYZ*, and *TMEM116*, which were implicated in autoimmunity.⁸⁷ Proteome-wide association

Table 2. Representative Multiomics Resources Relevant to Cardiovascular Research

Resources	Omics data	Samples (N)	Tissues	Cell types (N)	Population
GTEx	WGS, bulk RNA-seq	946	Multiple	Vary	Black, EAS, EUR, others
	WGS, snRNA-seq	16	Multiple	43	Not specified
Heart Cell Atlas	scRNA-seq, snRNA-seq, snATAC-seq, spatial	25	Heart	12	SA, EUR
KPMP	WGS, scRNA-seq, snRNA-seq, proteomics, spatial	812	Kidney	50+	Black, AS, EUR, Hispanic/Latino
METSIM	SNP array, WES, WGS, bulk RNA-seq, ATAC-seq, metabolomics; snRNAseq	84†	Adipose	12	EUR
OneK1K	SNP array, scRNA-seq	982	Blood	14	EUR
sc-eQTLGen	SNP array, scRNA-seq	247*	Blood	10+	EUR
scQTLbase	Bulk-tissue, scRNA-seq	2750*	Multiple	57	Black, EAS, EUR, admixed Peruvian
Single-Cell Atlas	scRNA-seq, snRNA-seq	22	Adipose	16	Not specified
SingleQ	scRNA-seq	3358*	Multiple	240+	Black, EAS, EUR, admixed Peruvian
STARNET	Bulk RNA-seq	850	Multiple	Vary	EUR
TOPMed	WGS, eQTL, sQTL	Varies	Multiple	Vary	Black, EAS, EUR, Hispanic/Latino

ATAC-seq indicates assay for transposase-accessible chromatin using sequencing; EAS, East Asian; EUR, European; GTEx, Genotype-Tissue Expression^{41,60}; Heart Cell Atlas⁶¹; KPMP, Kidney Precision Medicine Project⁶²; METSIM, METabolic Syndrome in Men^{63,64}; OneK1K⁶⁵; RNA-seq, RNA sequencing; sc-eQTLGen, single-cell eQTLGen Consortium⁵⁵; scQTLbase, integrated human single-cell eQTL portal⁵⁴; scRNA-seq, single-cell RNA-sequencing; Single-Cell Atlas⁶⁶; SingleQ, Single-cell Inter-active Genomic Library for eQTL⁵⁶; snRNA-seq, single-nuclei RNA- sequencing; STARNET, Stockholm-Tartu Atherosclerosis Reverse Networks Engineering Task⁶⁷; SA, South Asian; TOPMed, Trans-Omics for Precision Medicine^{40,59}; and WGS, whole-genome sequencing.

*Data curated from multiple studies.
†snRNAseq of subcutaneous adipose tissue.

studies and metabolome-wide association studies are still emerging approaches for hypertension and other traits relevant to cardiovascular risk. To date, proteomic and metabolomic data have been used primarily in Mendelian randomization settings in European ancestry populations or directly tested for association or admixture mapping with blood pressure or other cardiovascular risk factors in studies including admixed individuals.^{21,88,89} For example, Zhang et al⁸⁸ examined the association between cord blood metabolites and blood pressure traits in children and adolescents from the Boston Birth Cohort, uncovering metabolic associations with acylcarnitine and purine metabolites that could account for the early life causes of hypertension. The continued expansion of proteomic and metabolomic approaches, including proteome-wide association study and metabolome-wide association study efforts, into large, ancestrally diverse, and longitudinal settings will be critical for deepening mechanistic insight into hypertension and strengthening molecularly informed risk prediction.⁹⁰

LIMITATIONS OF MULTIOMIC APPROACHES

TWAS has limitations and caveats to consider when interpreting results.³⁸ A common misconception is that TWAS is designed to test causal genes. This approach helps identify candidate genes that are regulated by genetic variants and associated with outcomes, and follow-up analyses are still required to establish causality. Colocalization provides additional evidence

that a SNP in an expression model is functioning via gene expression regulation to affect the associated trait, and can be used to confirm associations identified by TWAS.^{91,92} It is also common to identify multiple TWAS-associated genes clustered in specific loci. The multiple-hit loci scenario might be due to false positives, largely driven by coregulation features (correlated gene expression across individuals; correlated predicted gene expression, which represents a small proportion of the total gene expression), linkage disequilibrium among a causal SNP influencing gene expression for a gene with another SNP that is an eQTL, and the influence of shared genetic variants across prediction models used for a pair of genes.³⁸ Furthermore, TWAS's effectiveness will be enhanced by utilizing gene expression data from tissues or cell types that are biologically relevant to the outcome.

An important challenge in genomic studies of admixed individuals is the limited availability of samples for replication. Populations originating from multiple ancestral sources may vary in the proportions of their ancestry and population structure across regions due to unique historical and migration events. This heterogeneity and the ongoing underrepresentation of some populations in genetic research create challenges for validation and broader application.

ARTICLE INFORMATION

Affiliations

Department of Genetics, School of Medicine (A.R.V.R.H., N.F.) and Department of Epidemiology, Gillings School of Global Public Health (N.F.), University of North Carolina, Chapel Hill.

Acknowledgments

The authors thank Kaylia May Reynolds, Wanbing Wang, and Wanyun Shi for assistance with figures and tables.

Sources of Funding

Funding for this research is provided by the National Institutes of Health R01-HG013163, R01-HL163972, and K26-DK138425 to N. Franceschini.

Disclosures

None.

REFERENCES

- Xue JL, Eggers PW, Agodoa LY, Foley RN, Collins AJ. Longitudinal study of racial and ethnic differences in developing end-stage renal disease among aged medicare beneficiaries. *J Am Soc Nephrol*. 2007;18:1299–1306. doi: 10.1681/ASN.2006050524
- Glantz NM, Duncan I, Ahmed T, Fan L, Reed BL, Kalirai S, Kerr D. Racial and ethnic disparities in the burden and cost of diabetes for US medicare beneficiaries. *Health Equity*. 2019;3:211–218. doi: 10.1089/heap.2019.0004
- Palaniappan LP, Allen NB, Almarazooq ZI, Anderson CAM, Arora P, Avery CL, Baker-Smith CM, Bansal N, Currie ME, Earlie RS, et al; American Heart Association Council on Epidemiology and Prevention Statistics Committee and Stroke Statistics Committee. 2026 heart disease and stroke statistics: a report of US and global data from the American Heart Association. *Circulation*. 2026;153:e275–e906. doi: 10.1161/CIR.0000000000001412
- National Center for Health Statistics. *National Health Interview Survey. Percentage of diagnosed diabetes for adults aged 18 and over, United States, 2019–2024*. National Health Interview Survey. Accessed 11/20/2025. https://www.cdc.gov/NHISDataQueryTool/SHS_adult/index.html
- Rowley WR, Bezold C, Arikian Y, Byrne E, Krohe S. Diabetes 2030: insights from yesterday, today, and future trends. *Popul Health Manag*. 2017;20:6–12. doi: 10.1089/pop.2015.0181
- Genovese G, Friedman DJ, Ross MD, Lecordier L, Uzureau P, Freedman BI, Bowden DW, Langefeld CD, Oleksyk TK, Uscinski Knob AL, et al. Association of trypanolytic ApoL1 variants with kidney disease in African Americans. *Science*. 2010;329:841–845. doi: 10.1126/science.1193032
- Sirugo G, Williams SM, Tishkoff SA. The missing diversity in human genetic studies. *Cell*. 2019;177:26–31. doi: 10.1016/j.cell.2019.02.048
- Mills MC, Rahal C. The GWAS diversity monitor tracks diversity by disease in real time. *Nat Genet*. 2020;52:242–243. doi: 10.1038/s41588-020-0580-y
- Cerezo M, Sollis E, Ji Y, Lewis E, Abid A, Bircan KO, Hall P, Hayhurst J, Jern S, Mosaku A, et al. The NHGRI-EBI GWAS catalog: standards for reusability, sustainability and diversity. *Nucleic Acids Res*. 2025;53:D998–D1005. doi: 10.1093/nar/gkae1070
- Adeyemo AA, Shriner D, Bentley AR, Gbadegesin RA, Rotimi CN. Evolutionary genetics and acclimatization in nephrology. *Nat Rev Nephrol*. 2021;17:827–839. doi: 10.1038/s41581-021-00483-7
- Sigma Type 2 Diabetes Consortim, Williams AL, Jacobs SB, Moreno-Macias H, Huerta-Chagoya A, Churchhouse C, Marquez-Luna C, Garcia-Ortiz H, Gomez-Vazquez MJ, Burr NP, et al. Sequence variants in SLC16A11 are a common risk factor for type 2 diabetes in Mexico. *Nature*. 2014;506:97–101. doi: 10.1038/nature12828
- Crawford JE, Amaru R, Song J, Julian CG, Racimo F, Cheng JY, Guo X, Yao J, Ambale-Venkatesh B, Lima JA, et al. Natural selection on genes related to cardiovascular health in high-altitude adapted andeans. *Am J Hum Genet*. 2017;101:752–767. doi: 10.1016/j.ajhg.2017.09.023
- Fumagalli M, Moltke I, Grarup N, Racimo F, Bjerregaard P, Jorgensen ME, Korneliusen TS, Gerbault P, Skotte L, Linneberg A, et al. Greenlandic Inuit show genetic signatures of diet and climate adaptation. *Science*. 2015;349:1343–1347. doi: 10.1126/science.aab2319
- Kramer HJ, Stilp AM, Laurie CC, Reiner AP, Lash J, Daviglus ML, Rosas SE, Ricardo AC, Tayo BO, Flessner MF, et al. African ancestry-specific alleles and kidney disease risk in Hispanics/Latinos. *J Am Soc Nephrol*. 2017;28:915–922. doi: 10.1681/ASN.2016030357
- Naik RP, Derebail VK, Grams ME, Franceschini N, Auer PL, Peloso GM, Young BA, Lettre G, Peralta CA, Katz R, et al. Association of sickle cell trait with chronic kidney disease and albuminuria in African Americans. *JAMA*. 2014;312:2115–2125. doi: 10.1001/jama.2014.15063
- Tishkoff SA, Verrelli BC. G6PD deficiency and malarial resistance in humans: insights from evolutionary genetic analyses. In: Dronamraju KR, ed. *Infectious Disease and Host-Pathogen Evolution*. Cambridge University Press. 2004;39–74.
- Reich D, Nalls MA, Kao WH, Akylbekova EL, Tandon A, Patterson N, Mullikin J, Hsueh WC, Cheng CY, Coresh J, et al. Reduced neutrophil count in people of African descent is due to a regulatory variant in the Duffy antigen receptor for chemokines gene. *PLoS Genet*. 2009;5:e1000360. doi: 10.1371/journal.pgen.1000360
- Rao S, Segar MW, Bress AP, Arora P, Vongpatanasin W, Agusala V, Essien UR, Correa A, Morris AA, de Lemos JA, et al. Association of genetic west african ancestry, blood pressure response to therapy, and cardiovascular risk among self-reported black individuals in the Systolic Blood Pressure Reduction Intervention Trial (SPRINT). *JAMA Cardiol*. 2021;6:388–398. doi: 10.1001/jamacardio.2020.6566
- Friedman DJ, Pollak MR. APOL1 nephropathy: from genetics to clinical applications. *Clin J Am Soc Nephrol*. 2021;16:294–303. doi: 10.2215/CJN.15161219
- Shriner D. Overview of admixture mapping. *Curr Protoc*. 2023;3:e677. doi: 10.1002/cpz1.677
- Reynolds KM, Horimoto A, Lin BM, Zhang Y, Kumiansyah N, Yu B, Boerwinkle E, Qi Q, Kaplan R, Daviglus M, et al. Ancestry-driven metabolic variation provides insights into disease states in admixed populations. *Genome Med*. 2023;15:52. doi: 10.1186/s13073-023-01209-z
- Horimoto A, Xue D, Cai J, Lash JP, Daviglus ML, Franceschini N, Thornton TA. Genome-wide admixture mapping of estimated glomerular filtration rate and chronic kidney disease identifies European and African ancestry-of-origin loci in hispanic and latino individuals in the United States. *J Am Soc Nephrol*. 2022;33:77–87. doi: 10.1681/ASN.2021050617
- Kopp JB, Smith MW, Nelson GW, Johnson RC, Freedman BI, Bowden DW, Oleksyk T, McKenzie LM, Kajiyama H, Ahuja TS, et al. MYH9 is a major-effect risk gene for focal segmental glomerulosclerosis. *Nat Genet*. 2008;40:1175–1184. doi: 10.1038/ng.226
- Kao WH, Klag MJ, Meoni LA, Reich D, Berthier-Schaad Y, Li M, Coresh J, Patterson N, Tandon A, Powe NR, et al; Family Investigation of Nephropathy and Diabetes Research Group. MYH9 is associated with nondiabetic end-stage renal disease in African Americans. *Nat Genet*. 2008;40:1185–1192. doi: 10.1038/ng.232
- Kopp JB, Yoshida T. Phenotypes of APOL1 high-risk status subjects. *J Am Soc Nephrol*. 2023;34:735–736. doi: 10.1681/ASN.0000000000000123
- Friedman DJ, Pollak MR. In search of the mechanism of APOL1 kidney disease. *J Am Soc Nephrol*. 2024;35:815–817. doi: 10.1681/ASN.0000000000000368
- Egbuna O, Zimmerman B, Manos G, Fortier A, Chirieac MC, Dakin LA, Friedman DJ, Bramham K, Campbell K, Knebelmann B, et al; VX19-147-101 Study Group. Inaxaplin for proteinuric kidney disease in persons with two APOL1 variants. *N Engl J Med*. 2023;388:969–979. doi: 10.1056/NEJMoa2202396
- Yang C, Hallmark B, Chai JC, O'Connor TD, Reynolds LM, Wood AC, Seeds M, Chen YI, Steffen LM, Tsai MY, et al. Impact of Amerind ancestry and FADS genetic variation on omega-3 deficiency and cardio-metabolic traits in Hispanic populations. *Commun Biol*. 2021;4:918. doi: 10.1038/s42003-021-02431-4
- Chilton FH, Manichaikul A, Yang C, O'Connor TD, Johnstone LM, Blomquist S, Schembre SM, Sergeant S, Zec M, Tsai MY, et al. Interpreting clinical trials with omega-3 supplements in the context of ancestry and FADS genetic variation. *Front Nutr*. 2021;8:808054. doi: 10.3389/fnut.2021.808054
- Marklund M, Wu JHY, Imamura F, Del Gobbo LC, Fretts A, de Goede J, Shi P, Tittle N, Wennberg M, Aslibekyan S, et al; Cohorts for Heart and Aging Research in Genomic Epidemiology (CHARGE) Fatty Acids and Outcomes Research Consortium (FORCE). Biomarkers of dietary omega-6 fatty acids and incident cardiovascular disease and mortality. *Circulation*. 2019;139:2422–2436. doi: 10.1161/CIRCULATIONAHA.118.038908
- Hellstrand S, Ericson U, Gullberg B, Hedblad B, Orho-Melander M, Sonestedt E. Genetic variation in FADS1 has little effect on the association between dietary PUFA intake and cardiovascular disease. *J Nutr*. 2014;144:1356–1363. doi: 10.3945/jn.114.192708
- Usova EI, Aliyeva AS, Yakovlev AN, Alieva MS, Prokhorikhin AA, Konradi AO, Shlyakhto EV, Magni P, Catapano AL, Baragetti A. Integrative analysis of multi-omics and genetic approaches—a new level in atherosclerotic cardiovascular risk prediction. *Biomolecules*. 2021;11:1597. doi: 10.3390/biom11111597
- Rasooly D, Giambartolomei C, Peloso GM, Dashti H, Ferolito BR, Golden D, Horimoto A, Pietzner M, Farber-Eger EH, Wells QS, et al. Large-scale multi-omics identifies drug targets for heart failure with reduced and preserved ejection fraction. *Nat Cardiovasc Res*. 2025;4:293–311. doi: 10.1038/s44161-025-00609-1
- Zhang J, Zhao H. eQTL studies: from bulk tissues to single cells. *J Genet Genomics*. 2023;50:925–933. doi: 10.1016/j.jgg.2023.05.003

35. Liu C, Joeanes R, Ma J, Wang Y, Sun X, Keshawar A, Sooda M, Huan T, Hwang SJ, Bui H, et al. Whole genome DNA and RNA sequencing of whole blood elucidates the genetic architecture of gene expression underlying a wide range of diseases. *Sci Rep*. 2022;12:20167. doi: 10.1038/s41598-022-24611-w
36. Luningham JM, Chen J, Tang S, De Jager PL, Bennett DA, Buchman AS, Yang J. Bayesian genome-wide TWAS method to leverage both cis- and trans-eQTL information through summary statistics. *Am J Hum Genet*. 2020;107:714–726. doi: 10.1016/j.ajhg.2020.08.022
37. Dai Q, Zhou G, Zhao H, Vosa U, Franke L, Battle A, Teumer A, Lehtimäki T, Raitakari OT, Esko T, et al; eQTLGen Consortium. OTTERS: a powerful TWAS framework leveraging summary-level reference data. *Nat Commun*. 2023;14:1271. doi: 10.1038/s41467-023-36862-w
38. Wainberg M, Sinnott-Armstrong N, Mancuso N, Barbeira AN, Knowles DA, Golan D, Ermel R, Ruusalepp A, Quertermous T, Hao K, et al. Opportunities and challenges for transcriptome-wide association studies. *Nat Genet*. 2019;51:592–599. doi: 10.1038/s41588-019-0385-z
39. Mai J, Lu M, Gao Q, Zeng J, Xiao J. Transcriptome-wide association studies: recent advances in methods, applications and available databases. *Commun Biol*. 2023;6:899. doi: 10.1038/s42003-023-05279-y
40. Orchard P, Blackwell TW, Kachuri L, Castaldi RJ, Cho MH, Christenson SA, Durda P, Gabriel S, Hersh CP, Huntsman S, et al; NHLBI TOPMed Consortium Multi-Omics Working Group. Cross-cohort analysis of expression and splicing quantitative trait loci in TOPMed. *medRxiv*. Preprint posted online February 21, 2025. doi: 10.1101/2025.02.19.25322561
41. GTEx Consortium. The GTEx Consortium atlas of genetic regulatory effects across human tissues. *Science*. 2020;369:1318–1330. doi: 10.1126/science.aaz1776
42. Kang JB, Raveane A, Nathan A, Soranzo N, Raychaudhuri S. Methods and insights from single-cell expression quantitative trait loci. *Annu Rev Genomics Hum Genet*. 2023;24:277–303. doi: 10.1146/annurev-genom-101422-100437
43. Franks A, Airoldi E, Slavov N. Post-transcriptional regulation across human tissues. *PLoS Comput Biol*. 2017;13:e1005535. doi: 10.1371/journal.pcbi.1005535
44. Brandes N, Linial N, Linial M. PWAS. Proteome-wide association study-linking genes and phenotypes by functional variation in proteins. *Genome Biol*. 2020;21:173. doi: 10.1186/s13059-020-02089-x
45. Kiessling P, Kuppe C. Spatial multi-omics: novel tools to study the complexity of cardiovascular diseases. *Genome Med*. 2024;16:14. doi: 10.1186/s13073-024-01282-y
46. Kuppe C, Ramirez Flores RO, Li Z, Hayat S, Levinson RT, Liao X, Hannani MT, Tanevski J, Wunnemann F, Nagai JS, et al. Spatial multi-omic map of human myocardial infarction. *Nature*. 2022;608:766–777. doi: 10.1038/s41586-024-05060-x
47. Linna-Kuusanen S, Schmauch E, Galani K, Ojanen J, Boix CA, Ord T, Toropainen A, Singha PK, Moreau PR, Harju K, et al. Transcriptomic and spatial dissection of human ex vivo right atrial tissue reveals proinflammatory microvascular changes in ischemic heart disease. *Cell Rep Med*. 2024;5:101556. doi: 10.1016/j.xcrm.2024.101556
48. Clair G, Soloyan H, Cravedi P, Angeletti A, Salem F, Al-Rabadi L, De Filippo RE, Da Sacco S, Lemley KV, Sedrakyan S, et al. The spatially resolved transcriptome signatures of glomeruli in chronic kidney disease. *JCI Insight*. 2024;9:e165515. doi: 10.1172/jci.insight.165515
49. Lin BM, Wen J, Horimoto A, Zhou L, Huang S, Dobre M, Go AS, Li Y, Franceschini N, Investigators CS. Transcriptome-wide association study of cardiovascular outcomes in chronic kidney disease: the chronic renal insufficiency cohort. *HGG Adv*. 2026;7:100562. doi: 10.1016/j.xhgg.2026.100562
50. Geoffroy E, Gregga I, Wheeler HE. Population-matched transcriptome prediction increases TWAS discovery and replication rate. *iScience*. 2020;23:101850. doi: 10.1016/j.isci.2020.101850
51. Head ST, Dai Q, Schildkraut J, Cutler DJ, Yang J, Epstein MP. CADET: Enhanced transcriptome-wide association analyses in admixed samples using eQTL summary data. *Am J Hum Genet*. 2025;112:1580–1596. doi: 10.1016/j.ajhg.2025.05.010
52. Breeze CE, Beck S, Berndt SI, Franceschini N. The missing diversity in human epigenomic studies. *Nat Genet*. 2022;54:737–739. doi: 10.1038/s41588-022-01081-4
53. Kachuri L, Mak ACY, Hu D, Eng C, Huntsman S, Elhawary JR, Gupta N, Gabriel S, Xiao S, Keys KL, et al. Gene expression in African Americans, Puerto Ricans and Mexican Americans reveals ancestry-specific patterns of genetic architecture. *Nat Genet*. 2023;55:952–963. doi: 10.1038/s41588-023-01377-z
54. Ding R, Wang Q, Gong L, Zhang T, Zou X, Xiong K, Liao Q, Plass M, Li L. scQTLbase, an integrated human single-cell eQTL database. *Nucleic Acids Res*. 2024;52:D1010–D1017. doi: 10.1093/nar/gkad781
55. van der Wijst M, de Vries DH, Groot HE, Trynka G, Hon CC, Bonder MJ, Stegle O, Nawijn MC, Idaghdour Y, van der Harst P, et al. The single-cell eQTLGen consortium. *Elife*. 2020;9:e52155. doi: 10.7554/eLife.52155
56. Zhou Z, Du J, Wang J, Liu L, Gordon MG, Ye CJ, Powell JE, Li MJ, Rao SS. a comprehensive database of single-cell expression quantitative trait loci (sc-eQTLs) cross human tissues. *Database (Oxford)*. 2024;2024:baae010. doi: 10.1093/database/baae010
57. Yang C, Gorijala P, Timsina J, Wang L, Liu M, Wang C, Brock W, Wang Y, Urano F, Sung YJ, et al. European and African ancestry-specific plasma protein-QTL and metabolite-QTL analyses identify ancestry-specific T2D effector proteins and metabolites. *Nat Commun*. 2025;16:7412. doi: 10.1038/s41467-025-62463-w
58. Cai Y, Johnson M, Haessler J, Molstad AJ, Hwang SJ, Joeanes R, Murabito JM, Tahir UA, Franceschini N, Gerszten RE, et al. Protein quantitative trait locus analysis in African American and non-Hispanic White individuals. *Genome Biol*. 2025;26:200. doi: 10.1186/s13059-025-03671-x
59. Taliun D, Harris DN, Kessler MD, Carlson J, Szpiech ZA, Torres R, Taliun SAG, Corvelo A, Gogarten SM, Kang HM, et al; NHLBI Trans-Omics for Precision Medicine (TOPMed) Consortium. Sequencing of 53,831 diverse genomes from the NHLBI TOPMed program. *Nature*. 2021;590:290–299. doi: 10.1038/s41586-021-03205-y
60. Eraslan G, Drokhlyansky E, Anand S, Fiskin E, Subramanian A, Slyper M, Wang J, Van Wittenberghe N, Rouhana JM, Waldman J, et al. Single-nucleus cross-tissue molecular reference maps toward understanding disease gene function. *Science*. 2022;376:eabl4290. doi: 10.1126/science.abl4290
61. Kanemaru K, Cranley J, Muraro D, Miranda AMA, Ho SY, Wilbrey-Clark A, Patrick Pett J, Polanski K, Richardson L, Litvinukova M, et al. Spatially resolved multiomics of human cardiac niches. *Nature*. 2023;619:801–810. doi: 10.1038/s41586-023-06311-1
62. de Boer IH, Alpers CE, Azeloglu EU, Balis UGJ, Barasch JM, Barisoni L, Blank KN, Bombardieri AS, Brown K, Dagher PC, et al; Kidney Precision Medicine Project. Rationale and design of the kidney precision medicine project. *Kidney Int*. 2021;99:498–510. doi: 10.1016/j.kint.2020.08.039
63. Jia H, Liu D. Integrative single-cell analysis of metabolic syndrome reveals novel cellular heterogeneity and differentiation dynamics in adipose tissue. *Diabetol Metab Syndr*. 2025;17:403. doi: 10.1186/s13098-025-01975-3
64. Laakso M, Kuusisto J, Stancakova A, Kuulasmaa T, Pajukanta P, Lusis AJ, Collins FS, Mohlke KL, Boehnke M. The Metabolic Syndrome in Men study: a resource for studies of metabolic and cardiovascular diseases. *J Lipid Res*. 2017;58:481–493. doi: 10.1194/jlr.O072629
65. Yazar S, Alquicira-Hernandez J, Wing K, Senabouth A, Gordon MG, Andersen S, Lu Q, Rowson A, Taylor TRP, Clarke L, et al. Single-cell eQTL mapping identifies cell type-specific genetic control of autoimmune disease. *Science*. 2022;376:eabf3041. doi: 10.1126/science.abf3041
66. Emont MP, Jacobs C, Essene AL, Pant D, Tenen D, Colleluori G, Di Vincenzo A, Jorgensen AM, Dashti H, Stefek A, et al. A single-cell atlas of human and mouse white adipose tissue. *Nature*. 2022;603:926–933. doi: 10.1038/s41586-022-04518-2
67. Koplev S, Seldin M, Sukhvasi K, Ermel R, Pang S, Zeng L, Bankier S, Di Narzo A, Cheng H, Meda V, et al. A mechanistic framework for cardiometabolic and coronary artery diseases. *Nat Cardiovasc Res*. 2022;1:85–100. doi: 10.1038/s44161-021-00009-1
68. Jones SC, Cardone KM, Bradford Y, Tishkoff SA, Ritchie MD. The impact of ancestry on Genome-wide Association Studies. *Pac Symp Biocomput*. 2025;30:251–267. doi: 10.1142/9789819807024_0019
69. Shahin T, Bolatbay D, Idaghdour Y. Toward global representation in single-cell genomics. *Trends Genet*. 2026;42:594–605. doi: 10.1016/j.tig.2026.02.008
70. Yang G, Mishra M, Perera MA. Multi-omics studies in historically excluded populations: the road to equity. *Clin Pharmacol Ther*. 2023;113:541–556. doi: 10.1002/cpt.2818
71. Franceschini N, Morris AP. Genetics of kidney traits in worldwide populations: the Continental Origins and Genetic Epidemiology Network (COGEN) Kidney Consortium. *Kidney Int*. 2020;98:35–41. doi: 10.1016/j.kint.2020.02.036
72. Lavange LM, Kalsbeek WD, Sorlie PD, Aviles-Santa LM, Kaplan RC, Barnhart J, Liu K, Giachello A, Lee DJ, Ryan J, et al. Sample design and cohort selection in the Hispanic Community Health Study/Study of Latinos. *Ann Epidemiol*. 2010;20:642–649. doi: 10.1016/j.jannepidem.2010.05.006
73. Denny JC, Rutter JL, Goldstein DB, Philippakis A, Smoller JW, Jenkins G, Dishman E; All of Us Research Program Investigators. The "all of us" research program. *N Engl J Med*. 2019;381:668–676. doi: 10.1056/NEJMs1809937
74. Giri A, Hellwege JN, Keaton JM, Park J, Qiu C, Warren HR, Torstenson ES, Kovesdy CP, Sun YV, Wilson OD, et al; Understanding

- Society Scientific Group. Trans-ethnic association study of blood pressure determinants in over 750,000 individuals. *Nat Genet.* 2019;51:51–62. doi: 10.1038/s41588-018-0303-9
75. Franceschini N, Fox E, Zhang Z, Edwards TL, Nalls MA, Sung YJ, Tayo BO, Sun YV, Gottesman O, Adeyemo A, et al; Asian Genetic Epidemiology Network Consortium. Genome-wide association analysis of blood-pressure traits in African-ancestry individuals reveals common associated genes in African and non-African populations. *Am J Hum Genet.* 2013;93:545–554. doi: 10.1016/j.ajhg.2013.07.010
 76. Liang J, Le TH, Edwards DRV, Tayo BO, Gaulton KJ, Smith JA, Lu Y, Jensen RA, Chen G, Yanek LR, et al. Single-trait and multi-trait genome-wide association analyses identify novel loci for blood pressure in African-ancestry populations. *PLoS Genet.* 2017;13:e1006728. doi: 10.1371/journal.pgen.1006728
 77. Liu Z, Shriner D, Hansen NF, Rotimi CN, Mullikin JC, Program NCS. Admixture mapping identifies genetic regions associated with blood pressure phenotypes in African Americans. *PLoS One.* 2020;15:e0232048. doi: 10.1371/journal.pone.0232048
 78. Zhu X, Cooper RS. Admixture mapping provides evidence of association of the VNN1 gene with hypertension. *PLoS One.* 2007;2:e1244. doi: 10.1371/journal.pone.0001244
 79. Sofer T, Baier LJ, Browning SR, Thornton TA, Talavera GA, Wassertheil-Smoller S, Daviglus ML, Hanson R, Kobes S, Cooper RS, et al. Admixture mapping in the Hispanic Community Health Study/Study of Latinos reveals regions of genetic associations with blood pressure traits. *PLoS One.* 2017;12:e0188400. doi: 10.1371/journal.pone.0188400
 80. Horimoto A, Sun Q, Lash JP, Daviglus ML, Cai J, Haack K, Cole SA, Thornton TA, Browning SR, Franceschini N. Admixture mapping of chronic kidney disease and risk factors in Hispanic/Latino individuals from Central America Country of Origin. *Circ Genom Precis Med.* 2024;17:e004314. doi: 10.1161/CIRCGEN.123.004314
 81. Kurniansyah N, Goodman MQ, Kelly TN, Elfassy T, Wiggins KL, Bis JC, Guo X, Palmas W, Taylor KD, Lin HJ, et al; NHLBI Trans-Omics in Precision Medicine (TOPMed) Consortium. A multi-ethnic polygenic risk score is associated with hypertension prevalence and progression throughout adulthood. *Nat Commun.* 2022;13:3549. doi: 10.1038/s41467-022-31080-2
 82. Onyenobi E, Zhong M, Soremekun O, Kamiza A, Boua R, Chikowore T, Group AR, Fatumo S, Choudhury A, Hazelhurst S, et al. Development and validation of polygenic risk scores for blood pressure traits in continental African Populations. *Circ Genom Precis Med.* 2025;18:e005048. doi: 10.1161/CIRCGEN.124.005048
 83. Feitosa MF, Kraja AT, Chasman DI, Sung YJ, Winkler TW, Ntalla I, Guo X, Franceschini N, Cheng CY, Sim X, et al; InterAct Consortium. Novel genetic associations for blood pressure identified via gene-alcohol interaction in up to 570K individuals across multiple ancestries. *PLoS One.* 2018;13:e0198166. doi: 10.1371/journal.pone.0198166
 84. Sung YJ, Winkler TW, de las Fuentes L, Bentley AR, Brown MR, Kraja AT, Schwander K, Ntalla I, Guo X, Franceschini N, et al; CHARGE Neurology Working Group. A large-scale multi-ancestry genome-wide study accounting for smoking behavior identifies multiple significant loci for blood pressure. *Am J Hum Genet.* 2018;102:375–400. doi: 10.1016/j.ajhg.2018.01.015
 85. Lee S, Miller CL, Bentley AR, Brown MR, Nagarajan P, Noordam R, Morrison JL, Schwander K, Westerman K, Kho M, et al; Biobank Japan Project. Large-scale blood pressure GWAS accounting for gene-depression interactions in 564,680 individuals from diverse populations. *HGG Adv.* 2026;7:100566. doi: 10.1016/j.xhgg.2026.100566
 86. Wang H, Noordam R, Cade BE, Schwander K, Winkler TW, Lee J, Sung YJ, Bentley AR, Manning AK, Aschard H, et al. Multi-ancestry genome-wide gene-sleep interactions identify novel loci for blood pressure. *Mol Psychiatry.* 2021;26:6293–6304. doi: 10.1038/s41380-021-01087-0
 87. Huang LJ, Zhang QX, Valenzuela RK, Xu JC, Yan F, Ma J. Identifying susceptibility genes for essential hypertension by transcriptome-wide association study. *Biochem Biophys Rep.* 2022;32:101387. doi: 10.1016/j.bbrep.2022.101387
 88. Zhang M, Brady TM, Buckley JP, Appel LJ, Hong X, Wang G, Liang L, Wang X, Mueller NT. Metabolome-wide association study of cord blood metabolites with blood pressure in childhood and adolescence. *Hypertension.* 2022;79:2806–2820. doi: 10.1161/HYPERTENSIONAHA.122.20139
 89. Reynolds KM, Lin BM, Armstrong ND, Ottosson F, Zhang Y, Williams AS, Yu B, Boerwinkle E, Thygarajan B, Daviglus ML, et al. Circulating metabolites associated with albuminuria in a Hispanic/Latino Population. *Clin J Am Soc Nephrol.* 2023;18:204–212. doi: 10.2215/CJN.09070822
 90. Corpas M, Pius M, Poburennaya M, Guio H, Dwek M, Nagaraj S, Lopez-Correa C, Popejoy A, Fatumo S. Bridging genomics' greatest challenge: the diversity gap. *Cell Genom.* 2025;5:100724. doi: 10.1016/j.xgen.2024.100724
 91. Giambartolomei C, Vukcevic D, Schadt EE, Franke L, Hingorani AD, Wallace C, Plagnol V. Bayesian test for colocalisation between pairs of genetic association studies using summary statistics. *PLoS Genet.* 2014;10:e1004383. doi: 10.1371/journal.pgen.1004383
 92. Hormozdiani F, van de Bunt M, Segre AV, Li X, Joo JWJ, Bilow M, Sul JH, Sankararaman S, Pasaniuc B, Eskin E. Colocalization of GWAS and eQTL signals detects target genes. *Am J Hum Genet.* 2016;99:1245–1260. doi: 10.1016/j.ajhg.2016.10.003